

Data Science and Analytics

1. Data Analysis Pipeline

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The Knowledge Discovery from Database Process (1)

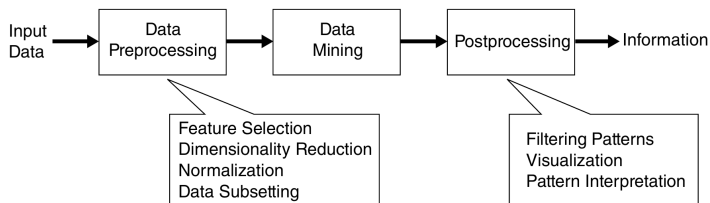
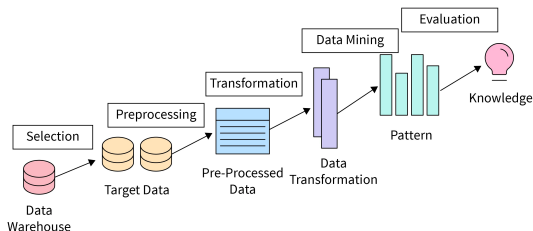


Figure 1: The process of knowledge discovery in databases (KDD)



The Knowledge Discovery from Database Process (2)

- KDD (Knowledge Discovery in Databases) is a **process of discovering useful knowledge and insights from large and complex datasets**.
- The KDD process involves a range of techniques and methodologies, including data preprocessing, data transformation, data mining, pattern evaluation, and knowledge representation.
- The KDD process **aims to identify hidden patterns, relationships, and trends in data** that can be used to make predictions, decisions, and recommendations.
- **KDD Process**
 - 1 **Data Selection** - The first step in the KDD process is identifying and selecting the relevant data for analysis. This involves choosing the relevant data sources, such as databases, data warehouses, and data streams, and determining which data is required for the analysis.
 - 2 **Data Preprocessing** - After selecting the data, the next step is data preprocessing. This step involves cleaning the data, removing outliers, and removing missing, inconsistent, or irrelevant data. This step is critical, as the data quality can significantly impact the accuracy and effectiveness of the analysis.

The Knowledge Discovery from Database Process (3)

- ③ **Data Transformation** - Once the data is preprocessed, the next step is to transform it into a format that data mining techniques can analyze. This step involves reducing the data dimensionality, aggregating the data, normalizing it, and discretizing it to prepare it for further analysis.
- ④ **Data Mining** - This is the heart of the KDD process and involves applying various data mining techniques to the transformed data to discover hidden patterns, trends, relationships, and insights. A few of the most common data mining techniques include clustering, classification, association rule mining, and anomaly detection.
- ⑤ **Pattern Evaluation** - After the data mining, the next step is to evaluate the discovered patterns to determine their usefulness and relevance. This involves assessing the quality of the patterns, evaluating their significance, and selecting the most promising patterns for further analysis.
- ⑥ **Knowledge Representation** - This step involves representing the knowledge extracted from the data in a way humans can easily understand and use. This can be done through visualizations, reports, or other forms of communication that provide meaningful insights into the data.

The Knowledge Discovery from Database Process (4)

- **Deployment** - The final step in the KDD process is to deploy the knowledge and insights gained from the data mining process to practical applications. This involves integrating the knowledge into decision-making processes or other applications to improve organizational efficiency and effectiveness.
- It is an iterative and interactive process.

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Structured and Unstructured Data (1)

1. Structured Data

- Data that is highly organized and follows a strict schema or predefined data model.
- Typically stored in tabular format consisting of rows (records) and columns (attributes).
- Each field has a fixed data type (e.g., integer, string, date).
- Easily processed using relational database management systems (RDBMS).
- Supports efficient querying using SQL (Structured Query Language).
- Commonly used in traditional business applications and enterprise systems.
- Enables fast retrieval, filtering, sorting, and aggregation of data.
- **Examples:**
 - Relational databases (MySQL, PostgreSQL, Oracle DB tables)
 - Spreadsheets (Microsoft Excel, Google Sheets)
 - Banking systems (account details, transactions, balances)
 - Student management systems (ID, name, grades, attendance)
 - Airline reservation systems (flight number, seat booking, schedule)
 - Inventory and supply chain databases

Structured and Unstructured Data (2)

2. Unstructured Data

- Data that does not follow any predefined format or organizational structure.
- Lacks a fixed schema, making it more complex to store and analyze.
- Often generated in large volumes from human activity and digital interactions.
- Requires advanced processing techniques such as Artificial Intelligence and Machine Learning.
- Cannot be easily stored in traditional row-column databases.
- Represents around 80–90% of total data generated today.
- **Examples:**
 - Textual data (emails, reports, PDFs, books, articles)
 - Multimedia content (images, audio recordings, videos)
 - Social media content (tweets, Facebook posts, Instagram captions)
 - Customer feedback and product reviews
 - Chat conversations and call center recordings
 - Surveillance videos and sensor-generated data

3. Semi-Structured Data

Structured and Unstructured Data (3)

- Contains some organizational properties but not in strict tabular form.
- Uses tags or markers to define structure.
- Easier to process than unstructured data but less rigid than structured data.
- **Examples:**
 - JSON and XML files
 - Emails (header + body)
 - NoSQL database records (MongoDB documents)
 - Web pages (HTML content)

Take Away:

- Structured data is easy to analyze but limited in flexibility.
- Unstructured data is highly rich in information but difficult to process.
- Semi-structured data provides a balance between flexibility and organization.

Structured and Unstructured Data (4)

Feature	Structured Data	Unstructured Data
Definition	Data that follows a fixed schema and organized format	Data that has no predefined format or structure
Organization	Highly organized in rows and columns	No specific organization or predefined model
Storage	Stored in relational databases (RDBMS)	Stored in data lakes, file systems, or NoSQL databases
Data Model	Requires predefined schema before storage	No fixed schema required (schema-on-read)
Processing	Easy to process using SQL and standard tools	Requires advanced techniques such as AI, ML, NLP, or computer vision
Querying	Fast, efficient, and straightforward querying	Complex and often requires specialized tools for analysis
Flexibility	Less flexible due to rigid structure	Highly flexible and adaptable to different formats
Examples	Databases, spreadsheets, financial records	Text, images, videos, emails, social media content
Data Volume	Smaller portion of total enterprise data	Majority of modern data (80–90%)

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Overview of Data Preprocessing (1)

Data Preprocessing

It is the process of transforming raw data into a clean and usable format suitable for data mining and machine learning tasks.

- Real-world data is often incomplete, noisy, inconsistent, and redundant, which can significantly reduce the performance of analysis algorithms.
- Data preprocessing is a critical step because poor-quality data leads to misleading or inaccurate analytical results.
- In practice, data preprocessing often consumes the majority of time in a data mining project.
- Proper preprocessing improves:
 - Data quality and consistency
 - Accuracy of mining and machine learning models
 - Computational efficiency
 - Interpretability of results

Overview of Data Preprocessing (2)

- **Major steps in data preprocessing:**

- **Data Cleaning:** Handles missing values, removes noise, and corrects inconsistencies in the data.
- **Data Integration:** Combines data from multiple sources such as databases, files, or web data into a unified dataset.
- **Data Transformation:** Converts data into appropriate formats using normalization, aggregation, or feature engineering techniques.
- **Data Reduction:** Reduces dataset size while preserving important information using sampling, dimensionality reduction, or compression techniques. There are a number of strategies for data reduction. These include **data aggregation** (e.g., building a data cube), attribute subset selection (e.g., removing irrelevant attributes through correlation analysis), **dimensionality reduction** (e.g., using encoding schemes such as minimum length encoding or wavelets), and **numerosity reduction** (e.g., “replacing” the data by alternative, smaller representations such as clusters or parametric models).
- **Data Discretization:** Converts continuous attributes into discrete intervals or categories for easier analysis.

Overview of Data Preprocessing (3)

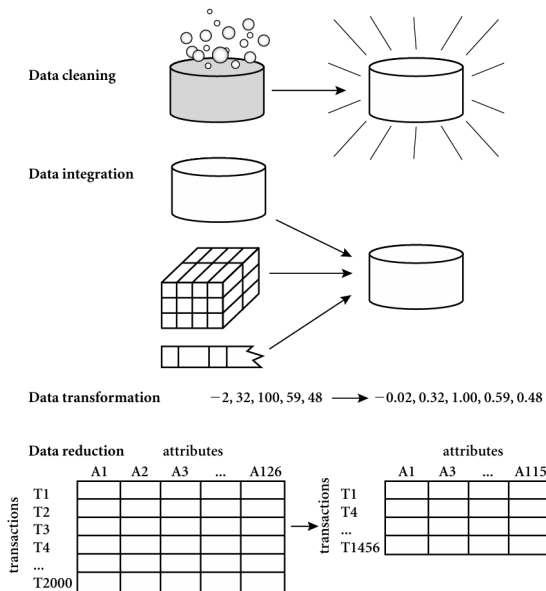


Figure 2: Forms of data preprocessing.

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Data Cleaning (1)

- Data Cleaning is the process of identifying and correcting (or removing) errors, inconsistencies, and inaccuracies in data.
- It is a crucial step in data preprocessing before analysis or machine learning.
- Real-world data is often incomplete, noisy, and inconsistent.
- Clean data improves accuracy, efficiency, and reliability of results.

Why Data Cleaning is Important?

- **Improves Data Quality:** Ensures correctness, completeness, and consistency of data.
- **Better Decision Making:** Clean data leads to accurate insights and reliable decisions.
- **Enhances Machine Learning Performance:** Models trained on clean data perform better.
- **Reduces Errors:** Prevents propagation of incorrect information in analysis.
- **Saves Time and Resources:** Reduces rework and debugging in later stages.
- **Ensures Data Consistency:** Standardized data is easier to integrate and use.

Data Cleaning (2)

Common Data Quality Issues

- **Missing Data:** Incomplete records or null values
- **Duplicate Data:** Repeated entries in dataset
- **Inconsistent Data:** Different formats or conflicting values
- **Noisy Data:** Errors or random variations in data
- **Outliers:** Extreme values that deviate from normal patterns

Techniques Used in Data Cleaning

- **Imputation Techniques:**
 - Mean, median, mode substitution
 - Regression-based imputation
- **Deletion Techniques:**
 - Row-wise deletion (remove incomplete records)
 - Column-wise deletion (remove features with many missing values)
- **Data Transformation:**
 - Normalization and scaling
 - Log transformation for skewed data

Data Cleaning (3)

- **Text Cleaning:**

- Lowercasing, removing punctuation, stopword removal
- Regex-based pattern cleaning

- **Duplicate Detection:**

- Exact matching and fuzzy matching techniques

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Data Integration (1)

Data Integration

It is the process of combining, sharing, or synchronizing data from multiple sources to provide users with a unified view.

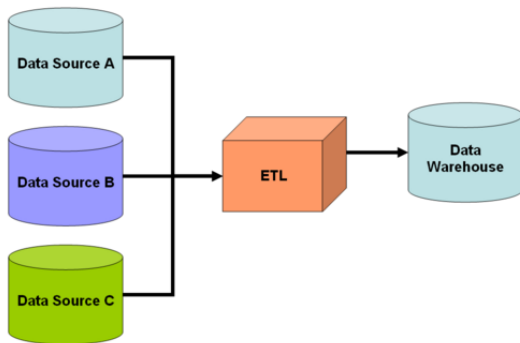


Figure 3: Simple schematic for a data warehouse. The **Extract, transform, load (ETL) process** extracts information from the source databases, transforms it and then loads it into the data warehouse.

Data Integration (2)

Data integration encourages collaboration between internal as well as external users. The data being integrated must be received from a heterogeneous database system and transformed to a single coherent data store that provides synchronous data across a network of files for clients.

- Combines data from multiple sources into a unified dataset.
- **Challenges:**
 - **Schema integration:** Combining different database schemas into a unified structure when data models, formats, or naming conventions differ.
 - **Entity identification:** Identifying and matching the same real-world entities (e.g., customers, products) across multiple data sources.
 - **Data redundancy:** Handling duplicate data that appears in multiple sources, which may lead to inconsistency and increased storage.
- **Techniques:**
 - **Data warehousing:** Storing integrated data in a centralized repository for efficient querying and analysis.
 - **Schema matching:** Aligning and mapping different data schemas to ensure compatibility and consistency.

Data Integration (3)

- **Conflict resolution:** Resolving inconsistencies in data values, formats, or meanings from different sources.
- **Ensures consistency and completeness of data.**



ETL Process in Data Integration

ETL (Extract, Transform, Load)

ETL is a fundamental process in data integration used to collect data from multiple sources, convert it into a consistent format, and store it in a centralized system such as a data warehouse.

Data Integration (4)

- **Extract:**

- Data is gathered from various sources such as databases, files, APIs, and cloud systems.
- Sources may have different formats and structures.

- **Transform:**

- Data is cleaned and converted into a consistent format.
- Includes removing duplicates, handling missing values, and applying business rules.

- **Load:**

- Transformed data is loaded into a target system such as a data warehouse.
- Data becomes ready for querying, reporting, and analysis.

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Data Transformation (1)

Data transformation techniques are methods used to convert raw data into a clean, structured, and usable format for analysis, modeling, or reporting

Data Transformation Methods

1 Normalization (Min-Max Scaling)

Rescales data into a fixed range, usually $[0,1]$.

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Example:

Suppose we have the dataset:

$$X = \{20000, 60000, 100000\}$$

Step 1: Identify minimum and maximum values

$$X_{min} = 20000, \quad X_{max} = 100000$$

Step 2: Compute normalized values

Data Transformation (2)

For $X = 20000$:

$$X' = \frac{20000 - 20000}{100000 - 20000} = 0.0$$

For $X = 60000$:

$$X' = \frac{60000 - 20000}{100000 - 20000} = \frac{40000}{80000} = 0.5$$

For $X = 100000$:

$$X' = \frac{100000 - 20000}{100000 - 20000} = 1.0$$

Result Table:

Salary	Normalized Salary
20000	0.0
60000	0.5
100000	1.0

Interpretation:

All values are now scaled between 0 and 1 while preserving their relative relationships.

Data Transformation (3)

2 Standardization (z-score Scaling)

Standardization transforms data so that it has mean 0 and standard deviation 1.

$$Z = \frac{X - \mu}{\sigma}$$

where:

- X = original value
- μ = mean of the dataset
- σ = standard deviation of the dataset

Example:

Suppose we have the dataset:

$$X = \{10, 12, 14, 16, 18\}$$

Step 1: Compute mean

$$\mu = \frac{10 + 12 + 14 + 16 + 18}{5} = 14$$

Data Transformation (4)

Step 2: Compute standard deviation

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

$$\sigma = \sqrt{\frac{(10 - 14)^2 + (12 - 14)^2 + \dots + (16 - 14)^2 + (18 - 14)^2}{5}} \approx 2.83$$

Step 3: Compute Z-score for $X = 16$

$$Z = \frac{16 - 14}{2.83} \approx 0.71$$

Interpretation: The value 16 is approximately 0.71 standard deviations above the mean.

Data Transformation (5)

3 Encoding Categorical Data

Encoding categorical data means converting non-numeric data into numeric format so that machine learning models can process it.

Why it is used: Machine learning models only understand numerical values, not text or categories.

Types of Encoding:

a) Label Encoding

Label encoding assigns a unique integer to each category.

Color	Encoding
Red	0
Blue	1
Green	2

Issue:

This method may introduce a false sense of ordering (e.g., $\text{Green} > \text{Blue} > \text{Red}$), which does not exist in reality.

Data Transformation (6)

b) One-Hot Encoding

One-hot encoding creates separate binary columns for each category.

Color	Red	Blue	Green
Red	1	0	0
Blue	0	1	0
Green	0	0	1

Advantage: No ordering bias is introduced.

Disadvantage: Increases dimensionality of the dataset.

❖ Aggregation

Aggregation refers to combining multiple records into summary values such as sum, average, count, minimum, or maximum.

What it is:

Aggregation transforms detailed (row-level) data into higher-level summarized information.

Why it is used:

It is used to analyze trends and patterns instead of working with raw, detailed data. This helps in decision-making and simplifies large datasets.

Data Transformation (7)

Common Aggregation Functions:

- **SUM** – total of values
- **AVG (Mean)** – average value
- **COUNT** – number of records
- **MIN** – smallest value
- **MAX** – largest value

Example:

Day	Sales
1	100
2	150
3	120
4	130
5	200

Data Transformation (8)

Step 1: Compute monthly total sales

$$\text{Total Sales} = 100 + 150 + 120 + 130 + 200 = 700$$

Step 2: Compute average daily sales

$$\text{Average Sales} = \frac{700}{5} = 140$$

Aggregated Result:

Metric	Value
Total Sales	700
Average Sales	140

Use Cases:

- Business dashboards for performance tracking
- Trend analysis over time (daily → monthly → yearly)
- Reporting systems in companies
- Data summarization in large databases

Data Transformation (9)

- Decision-making support in analytics

Interpretation:

Instead of analyzing individual transactions, aggregation provides a clear overview of overall performance and trends.

5 Smoothing

Smoothing is a data preprocessing technique used to remove noise and highlight meaningful patterns in data.

What it is:

It reduces random variation (noise) in data to make underlying trends more visible.

Why it is used:

Real-world data often contains fluctuations or outliers that can hide true patterns. Smoothing helps improve data quality for analysis.

Common Techniques:

- Moving Averages
- Bin Smoothing

Data Transformation (10)

1) Moving Average Smoothing:

In moving average, each value is replaced by the average of its neighboring values.

Example:

Raw data:

10, 12, 50, 11, 13

The value 50 is a spike (noise).

Step: Compute a simple smoothing effect using local averaging:

$$(10 + 12 + 50)/3 \approx 24$$

$$(12 + 50 + 11)/3 \approx 24.3$$

$$(50 + 11 + 13)/3 \approx 24.7$$

Smoothed data (conceptually):

$\approx 11, 24, 24, 12$

Data Transformation (11)

2) Bin Smoothing:

Data is divided into bins, and values in each bin are replaced using a representative value (mean, median, or boundary values).

Example:

$$\{10, 12, 50\} \rightarrow \text{bin mean} = 24$$

Why it matters:

- Improves trend detection in noisy datasets
- Useful in time-series forecasting
- Helps reduce the effect of outliers
- Makes data more stable for modeling

Interpretation:

Smoothing does not remove data but reduces random fluctuations so that underlying patterns become clearer.

Data Transformation (12)

6 Log Transformation

Log transformation is a data preprocessing technique that applies a logarithm function to reduce skewness in data.

What it is:

It transforms data using a logarithmic function to compress large values and make the distribution more symmetric.

Why it is used:

Many real-world datasets such as income, population, and sales are highly skewed. Log transformation helps stabilize variance and make patterns easier to analyze.

Formula:

$$X' = \log(X)$$

Effect of Log Transformation:

- Compresses large values
- Expands small values
- Reduces right-skewness in data

Data Transformation (13)

Example:

Income data:

1000, 100000

Applying log transformation (base 10 for simplicity):

$$\log(1000) = 3$$

$$\log(100000) = 5$$

Transformed data:

3, 5

Why it matters:

- Helps handle extreme values (outliers)
- Improves model performance for skewed data
- Common in financial and demographic datasets

Data Transformation (14)

Interpretation:

Log transformation reduces the impact of very large values, making the distribution more balanced and easier to analyze.

7 Generalization

Generalization refers to converting low-level (detailed) data into higher-level, more abstract data to reduce complexity and improve data usability.

What it is:

Generalization replaces specific data values with broader categories or concepts, making the dataset simpler and more meaningful for analysis.

Why it is used:

It reduces data granularity, helps discover patterns, and improves efficiency in data mining and analysis by avoiding unnecessary detail.

How it works:

Specific values are replaced with higher-level categories using concept hierarchies.

Data Transformation (15)

Example:

Original data (low-level):

- Kathmandu, Lalitpur, Bhaktapur

After generalization (city $\rightarrow$ region):

- Bagmati Province

Why it matters:

- Reduces data complexity
- Helps in pattern discovery
- Improves scalability of data analysis
- Protects privacy by hiding detailed information

Interpretation:

Generalization simplifies data by replacing detailed values with higher-level concepts, making it easier to analyze and understand large datasets.

Data Discretization (1)

Discretization is the process of converting continuous numerical data into discrete intervals (also called bins or categories).

- It transforms continuous attributes into a finite number of intervals, making the data easier to interpret and process.
- **Why it is used:** Many machine learning algorithms and data mining techniques perform better with categorical or simplified data. Discretization also helps reduce noise and improves interpretability.
- **Key Idea:** Instead of working with exact values, we group values into ranges.

Example:

Age data:

18, 22, 25, 30, 35, 40, 45

After discretization:

Age	Category
18–25	Young
26–40	Adult
41–60	Middle-aged

Data Discretization (2)

Common Methods of Discretization:

- **Binning:** Groups data into equal-width or equal-frequency intervals.
- **Histogram Analysis:** Uses the distribution of data to identify natural break-points or clusters.
- **Decision Tree-Based Methods:** Uses splitting points from decision trees (e.g., information gain, Gini index) to find optimal intervals.

Types of Binning:

- **Equal-width binning:** Each bin has the same range size.
- **Equal-frequency binning:** Each bin has the same number of data points.

Why it matters:

- Reduces complexity of continuous data
- Helps remove noise and minor fluctuations
- Improves performance of certain ML algorithms
- Makes data more interpretable for humans

Interpretation:

Discretization simplifies continuous data by converting it into meaningful intervals, making analysis easier and more efficient.